Problem Tracker — Segment 4: A Real Design Mistake, Fixed

4 of 8 · 5 min

Shows you can spot and correct your own design debt, not just design well the first time

Opening line: "I want to walk through a mistake I made and caught myself — not something a code reviewer found, not something that broke in production. That's a better story than pretending everything was right the first time."

The 5-Minute Run of Show

0:00–0:30	Framing (above)
0:30–2:30	The two-SQLite-files migration (headline)
2:30–4:00	A fresher one: the query-string duplicate bug
4:00–4:45	A third, quick one: the double-counted discount
4:45–5:00	The throughline + close

If They Only Remember One Thing

- All three of these were caught by *me* — by testing, by re-deriving a number instead of trusting it, or by a user (my own future self) asking "are you sure?" — not by someone else finding them first.

Section-by-Section Script

The two-SQLite-files migration (0:30–2:30) — headline

- *"The original version had two separate SQLite files — a scraper DB and a reviews DB. SQLite foreign keys can't span two files, so 'referential integrity' was whatever a Python dict-based join in a migration script happened to enforce. Nothing stopped an orphaned review row at the engine level."*
- *"Fix: one file, one problems table, a real enforced FK with ON DELETE CASCADE. Scope also widened from a 112-row 'completed only' subset to the full problem catalog in the process."*

The query-string duplicate bug (2:30–4:00) — this week

- *"I briefly widened the scraper to NeetCode's full ~973-problem catalog, then reverted it after real issues — pagination, slower runtime. When I checked the numbers afterward, the dashboard said 354 problems tracked. My own instinct was to trust it. The person I was demoing to asked 'are you sure that's accurate, no duplicates?' — and they were right to ask: it was actually 481, with 127 duplicate pairs."*
- *"Root cause: NeetCode's URL includes which list you were viewing — ?list=neetcode150 vs ?list=allNC — and url was my primary key, query string included. Switching scrape targets made the same problem get stored under two identities. Fixed the URL construction to strip the query string, then wrote a migration to merge the 127 existing duplicate pairs, re-pointing review history before deleting the redundant rows."*
- **Why this is a good story, not an embarrassing one:** *"I didn't defend the number — I verified it, found I was wrong, and fixed it with the same rigor as a fresh bug. That's the habit that matters more than never being wrong in the first place."*

A third, quick one: the double-counted discount (4:00–4:45)

- *"While writing unit tests for the scoring heuristic, I found it subtracted a 'cumulative mastery' discount twice when the most recent review happened to be 'Solved' — once in a branch specific to that outcome, once again unconditionally right after. Collapsed to a single discount, regression-tested so it can't silently come back. Caught by test-writing discipline on a function that already looked done, not by code review."*

The throughline (4:45–5:00): "Three different bugs, three different discovery methods — a schema review, a direct question from someone I was demoing to, and a test I wrote for an unrelated reason. None of them were found by someone else's code review. I try to build in enough places where being wrong is cheap to discover."